Performance Analysis of Massive MIMO Systems: Efficiency, Signal Processing, and Cell-Free Architectures

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Abstract—**The paper will provide an extensive overview of Massive Multiple-Input Multiple-Output (MIMO) technology, which is one of the enabling technologies of the future wireless communication networks in the fifth and subsequent generation. We investigate the fundamental advantages of implementing large-scale antenna arrays such as a large improvement in spectral and energy efficiency, and explain the channel models that govern the performance of the system. To demonstrate the trade-offs between computation complexity and network capacity, a comparative analysis of the linear signal processing algorithms are presented in both downlink precoding, including Maximum Ratio Transmission and Zero-Forcing, and uplink detection, including Minimum Mean Square Error and new deep learning methods. Moreover, we discuss the imperative tradeoff between spectral efficiency and energy efficiency optimization. The paper also examines the transformative Cell-Free Massive MIMO architecture as a way forward to the next-generation network, including how it can address cell-edge constraints by distributed access points. Lastly, we discuss the current issues in deployment, such as pilot contamination, bottlenecks in fronthaul capacity, and hardware limitations, and take a holistic view of the current and future of Massive MIMO systems.**

Index Terms—**Massive MIMO, 5G Networks, Spectral Efficiency, Energy Efficiency, Precoding, Detection Techniques, Cell-Free Architecture**

I. Background

The idea of massive multiple-input multiple-output (MIMO) wireless communications, including providing cellular base stations with a huge number of antennas, has proved the ability to allow spectrum and energy efficiency improvements of orders of magnitude with relatively simple (linear) processing [1]. In terms of MIMO concepts, new technologies have been used in fifth generation (5G) and above-5G communication systems [2].

II. Introduction

Every year, there is a significant increase in the number of mobile phone users. Users demand fast access to multimedia services and faster internet connection. Furthermore, the development of smart cities has grown to the level where a huge and varied set of devices distributed around the urban area produce enormous amounts of data that are required to be transmitted. So, higher energy efficiency, higher data rates, higher spectral efficiency, better mobility and larger network capacity are all required [3]. Massive MIMO (multiple-input multiple-output) has emerged as a new technology for wireless communication systems in the last few years. The ability to create extremely narrow signals for many different users by thousands of transmitting and receiving antennas has generated attention from academia and industry. The main idea is to increase the number of antennas at the base station by a minimum value of double order of magnitude. Massive MIMO makes it possible to transmit many different spatial signals to several different user devices in the same cell, at the same time and at the same frequency [4]. Lately, significant efforts in research have been done to improve the traditional transmission of multi-antenna to obtain high link performance and high spectrum efficiency [2]. MIMO has been a crucial technology that has been used since the release of the third generation (3G), and it has been used to improve the wireless transceivers' performance [3]. This paper will discuss the benefits, challenges, efficiency and channel Models of massive MIMO. In addition, the precoding techniques, detection techniques and cell-free massive MIMO.

III. Benefits of Massive MIMO

The elimination of the effects of uncorrelated noise and small-scale fading effects is one of the mathematical benefits of implementing large-scale antenna arrays. The random matrix theory states that a cell will require no energy to transmit a bit as the number of antennas becomes infinitely large and the number of users that can be served does not depend on the size of the cell. Also, the large amount of spatial degrees of freedom may be used strategically to form transmission signals in a hardware friendly way or to totally cancel inter-user interference [1]. The architecture radically changes the power relationship of the system through offering massive array gains. This means that the transmission power needed by user equipments can be cut drastically in proportion to the number of base station antennas with perfect channel state information (CSI). When CSI is not perfect, it is still possible to reduce the transmit power in direct proportion to the square root of the antenna count without performance loss. Alternatively, should power reduction not be the main goal, this basic gain can be reused to dramatically increase the communication range that is possible with traditional single-antenna networks [1]. At the same time the technology provides considerable multiplexing gains of the order of $\mathcal{O}(K)$, which means that the total system throughput is proportional to the number of served users. These extraordinary capacity and power-saving improvement can be achieved with the circuit power consumption extremely low. This is mostly because close-optimal throughput performance can be achieved by making use of the basic techniques like maximal-ratio combining in the uplink and maximal-ratio transmission in the downlink which effectively circumvents the prohibitively high computational requirements of the complex maximum likelihood detection or dirty paper coding algorithms [5].

IV. Channel Models

Massive MIMO depends on measuring the responses of the frequencies of the actual propagation channels, and to achieve that, the base station or the users send known training signals, and the receiver on the other side receives them and measures the frequency response. Training and data transmission in massive MIMO system are performed in a slot whose duration is determined so that no one transmits more than an amount of a wavelength during this slot [6]. System analysis and performance evaluation depends significantly on the channel model. One of the factors that significantly impacts the channel model is the configuration of the antenna array. Each measurement takes place based on a specific array. There are three main types of antenna configurations, which are 1D, 2D and 3D arrays. Each configuration and their sizes affect on the channel characteristics, so the massive MIMO has different performances based on the type of antenna configuration [4]. There are two main types of channel models, which are correlation-based and geometry-based stochastic channel models. Correlation-based models are mostly used for mathematical evaluations of system performance, while geometry-based stochastic channel models include those used for real-life wireless communication system assessments [4].

V. Signal Processing

The fifth generation (5G) and beyond 5G (B5G) communication systems depend heavily on massive multiple-input multiple-output (MIMO). Unfortunately, using a lot of antennas and radio frequency chains (RF) significantly increases the complexity of enormous MIMO systems. As a result, numerous studies have been carried out to determine the best precoding algorithm with the least amount of complexity. Hundreds of antennas on the base station are required to accommodate concurrent transmissions, which require effective signal processing algorithms. This subsection will describe the main linear algorithms of downlink precoding and uplink detection in terms of their performance and complexity trade-offs [2].

A. *Downlink Precoding Techniques*

Massive MIMO systems have many users sharing time-frequency resources between the base station and the users. Downlink precoding is necessary to concentrate the energy of the signal being transmitted on targeted users and inter-user interference is reduced. The use of linear precoding has been widely used in real-life implementations because it offers an optimal trade-off between spectral efficiency and computational complexity relative to non-linear complex techniques as shown in the system model in Figure 1 [2].

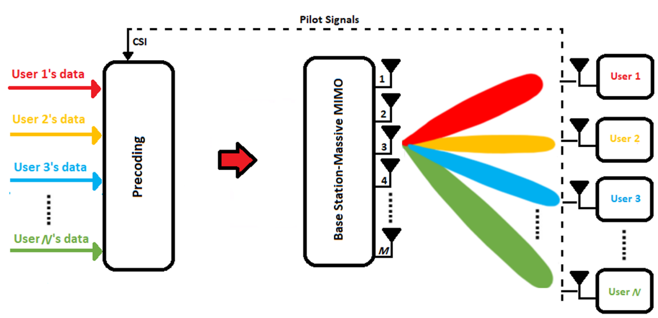

Fig. 1. System model of a downlink Massive MIMO system illustrating the functional position of the precoding block [2].

Maximum Ratio Transmission (MRT)

MRT, which is conceptually identical to Matched Filter (MF) used in uplink detection, is the simplest linear precoding scheme. The main aim of MRT is to achieve the maximum received signal-to-noise ratio (SNR) of the desired user by aligning the transmitted signal with the conjugate of the channel vector [7]. The MRT precoding matrix is mathematically:

$$P_{\text{MRT}} = \sqrt{\beta}(H^*) \tag{1}$$

where β is a power scaling factor, and H^* is the complex conjugate of the channel matrix H. MRT is computationally very efficient, but it inherently does not take into account multi-user interference. One benefit of conjugate beamforming and matched filtering is the ability to process massive MIMO signals locally at each antenna [6]. But the channel vectors of any two users become orthogonal when the number of base station antennas is raised to an exceptionally large volume. This inherently inhibits interference enabling MRT to effectively perform in extreme Massive MIMO configurations [2], [7]. From an energy efficiency point of view, MF precoding is appealing because it has low computational complexity and lower processing needs [7], [8].

Zero-Forcing (ZF) Precoding

Zero-Forcing precoding, in contrast to MRT, actively reduces multi-user interference. ZF does it by sending the transmitted signal to the null space of the interference channels effectively eliminating inter-user interference altogether [2]. It can be done by computing the pseudo-inverse of the channel matrix. ZF precoding matrix can be represented as:

$$P_{\text{ZF}} = \sqrt{\beta}H^*(G^{-1}) \tag{2}$$

where $G = H^T H^*$ is the Gram matrix. This results in very high spectral efficiency and capacity at ZF compared to MRT, especially in interference-limited scenarios or where the number of antennas is not large. Zero forcing may outperform conjugate beamforming and matched filtering under high SNR conditions [6]. Also, ZF can lower energy efficiency compared to MF, especially in systems with many users or in less than perfect channel conditions [6]. The trade-off, however, is that it is more computationally complex since heavy operations of matrix inversion are needed at the base station [2], [9]. According to [7], [8], The decision between MF and ZF depends on whether the system values energy savings or data rate performance more. While MF is more energy-efficient because it is simpler, ZF offers higher spectral efficiency by reducing interference. However, practical evaluations have shown that the performance gain of ZF over MF is marginal under realistic channel models, and therefore the additional complexity of ZF is not always justified.

B. Detection

During the uplink stage, the base station should be able to effectively demodulate and decode the overlapping signals of multiple users who send signals at the same time through a single antenna. This involves having strong symbol detection algorithms to deal with the high dimensionality of channel matrix. Although the best non-linear detectors, including Maximum Likelihood (ML) offer better performance, their computational complexity increases exponentially and therefore is impractical to be computed in a Massive MIMO implementation. As a result, linear detection methods are highly embraced because they offer a good trade-off between performance and computation overhead [3].

Matched Filter (MF) Detection

MF treats the contribution of other sub-streams as noise only [3]. The received signal computed based on MF is estimated as:

$$\hat{x}_{\text{MF}} = \mathcal{S}(H^H y) \tag{3}$$

with slicer function, $\mathcal{S}(.)$, and conjugate transpose of the channel matrix, H^H [3], [10]. Although computationally very efficient, MF fully neglects the multi-user interference, thus it can only be useful when the number of antennas is truly enormous [3].

Zero-Forcing (ZF) Detection

The ZF is superior to the MF detector in that it tries to maximize the ratio of the received signal to the interference and noise (SINR) [3]. The ZF mechanism relies on the inversion of the channel matrix H to cancel out its impact, by means of the equalization matrix:

$$A_{\text{ZF}}^H = (H^H H)^{-1} H^H = H^{\dagger} \tag{4}$$

and H^+ is the pseudo-inverse of Moore-Penrose. Though ZF works remarkably well in the interference limited conditions, it experiences a large noise amplification in the ill-conditioned channel matrix cases [3].

Minimum Mean Square Error (MMSE) Detection

To overcome the noise augmentation of ZF, the MMSE algorithm takes into consideration the noise effect on the equalization process. The signal-vector transmitted is estimated by:

$$\hat{s} = (H^H H + \sigma^2 I_{2K})^{-1} H^H y = W^{-1} \hat{y} \tag{5}$$

In which, the identity matrix is denoted as I_{2K} and $W = G + \sigma^2 I_{2K}$ is the MMSE filtering matrix based on the Gram matrix, $G = H^H H$. The algorithm to invert the matrix W^{-1} directly has a complexity of the order of $\mathcal{O}(K^3)$ [11]. In order to prevent this, more complex algorithms such as the Richardson method may be used to lower the complexity to the level of $\mathcal{O}(K^2)$ [11].

Deep Learning-Based Detection

The latest developments are based on the limitations and excessive computational power of traditional signal processing, where artificial intelligence is employed to simplify the detection algorithms [12]. Traditional iterative detectors, including Approximate Message Passing (AMP)-based and Orthogonal AMP (OAMP)-based iterative detectors, provide near-optimal performance, but can be costly to compute.

In response to it, unfolding iterative algorithms have been converted into a deep neural network format to form model-driven deep learning networks, including the OAMP-Net, in response to it [12]. This architecture replaces the black box approach of the neural network with a small number of learnable scalar parameters to optimize mean and variance update step sizes in the MMSE denoiser on-the-fly. The major benefit of such a solution is its ability to work well with time-varying channels with a single training phase, with minimal online computation delay. Moreover, since only the number of trainable variables depends on the number of network layers and does not depend on the number of antennas, the training process is very efficient and fast. Comprehensive analyses show that this model-based network can enhance the Bit Error Rate (BER) performance significantly over both independent Rayleigh and spatially correlated Massive MIMO channels when using conventional iterative algorithms and model-based networks [12].

VI. System Efficiency

Urgently required is the paradigm shift to Massive Multiple-Input Multiple-Output (MIMO) architecture to jointly optimize Spectral Efficiency (SE) and Energy Efficiency (EE) to cater to the larger and more challenging-to-satisfy data rate requirements and sustainability needs of modern wireless networks [13]. Although SE is considerably improved with the deployment of hundreds of antennas by aggressive spatial multiplexing, it also presents a larger footprint in terms of circuit power consumption. Therefore, the complexity of the interaction of these two crucial metrics is of primary importance to understand the interaction of both metrics with each other and their influence on each other respectively [9]. This part is a thorough evaluation of SE and EE which explicitly looks at the inherent trade-offs of co-located Massive MIMO systems. We explore the interaction between transmit power, number of base station antennas, hardware dissipation and determine the theoretical limits and optimal operating regimes of efficiency in the overall system by exploiting rigorous analytical models, thus defining the theoretical limits and optimal operating regimes of the entire system [7].

A. Spectral Efficiency (SE)

Spectral Efficiency is used to measure the system throughput within a bandwidth. Massive MIMO systems enhance SE dramatically through an array size (i.e., the base station antenna number) of M being vastly bigger than the number of active, single-antenna users (i.e., number of Users, K). The effects of channel fading and noise, the effect of noise and fading vanishes as M becomes infinitely large [7]. Simple linear signal processing can be applied in this phenomenon to ensure that the base station can serve many users sharing a common time-frequency resource. Most importantly, mathematical proofs reveal that perfect Channel State Information (CSI) means that the transmit power per user could be reduced by a constant scaling factor of $1/M$ without reducing the spectral efficiency that is achievable [7]. Although imperfect CSI still allows the possibility of reducing the transmit power in direct proportion to $1/\sqrt{M}$ to enable high-data rates, the radiated power is also substantially decreased, as shown in Figure 2 [7].

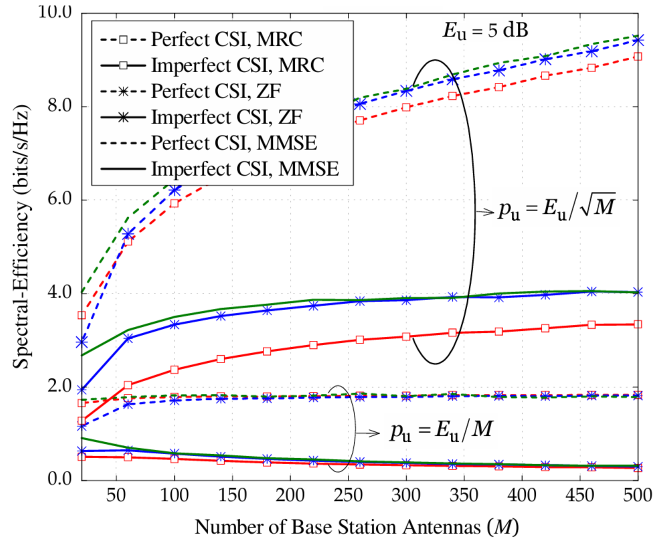

Fig. 2. Sum rate vs. number of antennas (M) for different linear processing techniques [7].

B. Energy Efficiency (EE) and Power Consumption Models

While SE focuses on capacity, whereas Energy Efficiency (usually expressed in bits/Joule) plays a decisive role in sustainable network operations. The myth is that the number of antennas scaled up drastically raises power consumption. But proper EE modeling should incorporate the overall power consumption that comprises of the transmit power, as well as the circuit power taken by the electronic hardware components [14]. System models demonstrate that there exists a single, globally optimal number of base station antennas and transmit power that results in the greatest energy efficiency of the system [14]. Moreover, the fact that realistic power consumption models that describe hardware impairments and circuit overhead are used supports the fact that indeed maximum energy efficiency is attained by deploying hundreds of antennas to serve a fairly large number of users. This conclusively demonstrates that Massive MIMO is also an energy-efficient (green) technology in the right design as explained in the power breakdown in Figure 2 [9].

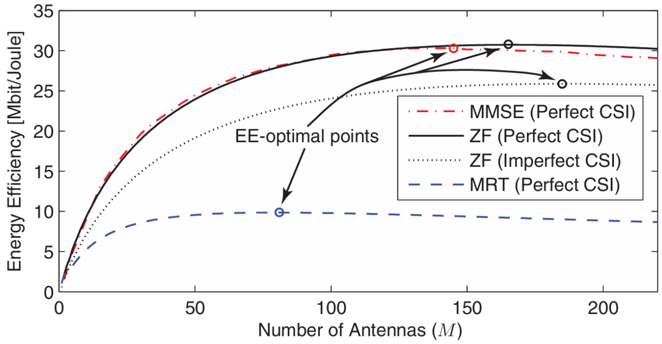

Fig. 3. Maximal EE for different number of BS antennas and different processing schemes in the single-cell scenario [9].

C. The SE-EE Trade-off

Although SE and EE have improved on an individual basis, a trade-off between the maximization of both measures in wireless communications still exists. The theoretical trade-off curve between SE and EE is formed by extensive surveys on this dynamic. Achieving a high Spectral Efficiency of a network usually demands large transmission powers, which essentially worsens the Energy Efficiency. On the other hand, it frequently decreases achievable data rates to ensure a strictly maximum EE. Thus, maximization of a single parameter in isolation is not the final objective in Massive MIMO deployment, but to develop optimization algorithms that identify the operating point on this SE-EE trade-off curve given the particular network demands [13].

VII. Cell-Free Massive MIMO: The Pathway to 6G

In order to address the intrinsic drawbacks of conventional cell-centric networks, including excessive inter-cell interference, and poor edge-user performance, a paradigm shift is needed. In this section, the transformative architecture of Cell-Free Massive MIMO is presented, discussing the main concepts behind it, its efficiency advantages, and the issues of its practical implementation [15].

A. The Cell-Free Concept

A Cell-Free Massive MIMO system is a system that is composed of many distributed access points (APs) and a few users at once. It is an architecture that removes the boundary of the cells, unlike a traditional cellular network. All distributed APs are connected to a central processing unit (CPU) via a high capacity backhaul network. Through such coordinated action like cooperating phase-coherently, this distributed version of the solution eliminates the Classic "cell-edge" problem, in which the network always offers equally good service to all users, no matter where they are physically located [15]. The system extensively depends on Time-Division Duplexing (TDD) and simple linear signal processing, including matched filtering, to work effectively and manage the huge number of antennas. Through the channel reciprocity, the APs are capable of precisely estimating the channel conditions in their immediate environment through uplink pilots by the users. This enables the Cell-Free architecture to be highly scalable and reliable as this mechanism minimizes signalling overhead and enables the network to average out multi-user interference Without extra complexity [15].

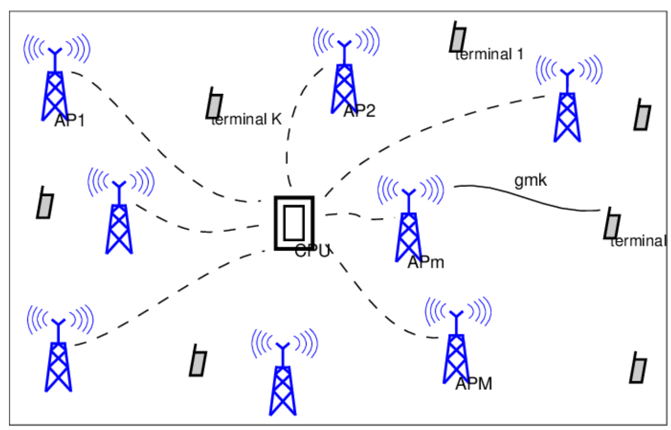

Fig. 4. Cell-Free Massive MIMO system. [15].

Mathematical System Model

using conjugate beamforming, assume that there are M distributed access points (APs) and K single-antenna users in a network. Equation (26) provides the received complex baseband signal at the k-th user in:

$$
\begin{aligned}
r_{d,k} =& \sqrt{\rho_d} \sum_{m=1}^{M} \sqrt{\eta_{mk}} g_{mk} \hat{g}_{mk}^{*} q_k \\
&+ \sqrt{\rho_d} \sum_{k' \neq k}^{K} \sum_{m=1}^{M} \sqrt{\eta_{mk'}} g_{mk} \hat{g}_{mk'}^{*} q_{k'} + w_{d,k}
\end{aligned}
\tag{6}
$$

- M is the total number of distributed APs.
- K is the total number of users (where $M \gg K$).
- $r_{d,k}$ is the received signal at the k-th user.
- ρ_d is the normalized maximum downlink transmit power.
- η_{mk} represents the power control coefficient between the m-th AP and the k-th user.
- g_{mk} is the actual channel coefficient between the m-th AP and the k-th user.
- $\hat{g}^*_{mk}$ denotes the channel estimate used by the AP for conjugate beamforming (precoding).
- q_k is the intended data symbol for the k-th user.
- $w_{d,k}$ is the additive noise at the receiver.

The first term in this equation is the desired signal that is to be coherently sent by the All the available APs of the M APs to the $k-th$ user, and the second term is the multi-user interference caused by the signal sent to co-channel users in the network [15].

Cell-Free vs. Cellular Massive MIMO

Unlike traditional Massive MIMO where all antennas are in one place, the Cell-Free architecture has a significantly higher 95%- likely per-user throughput. This is mainly because of the higher gain in macro-diversity whereby the users have a far greater likelihood of being within proximity of at least one AP. Although co-located systems are effective in areas close to the base station, the Cell-Free system provides a more equal distribution of data rates with max-min fairness power control and is therefore a strong contender in densely populated urban areas [15].

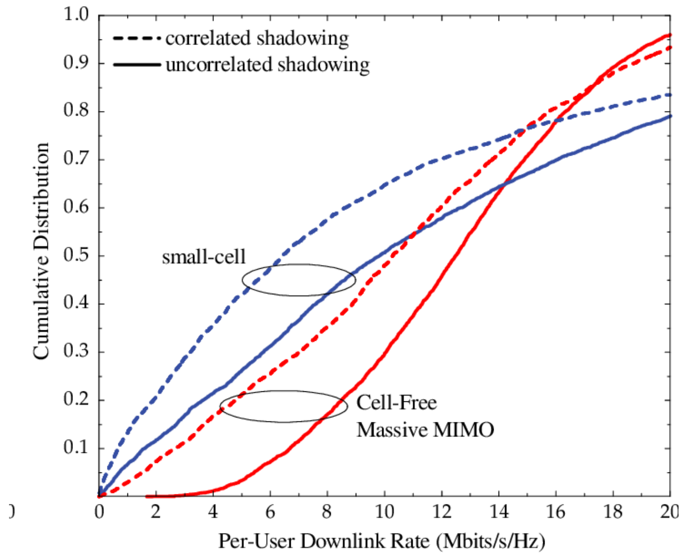

Fig. 5. Cell-Free Massive MIMO and conventional Small-Cell systems are compared using the cumulative distribution of the per-user throughput. [15].

Energy Efficiency and Sustainability

The Cell-Free architecture offers sustainability as a fundamental benefit, as it offers a route towards green communication and net-zero energy objectives. The proximity of the access points makes the distance between the transmitter and the receiver very short thus minimizing the transmit power needed but ensuring high spectral efficiency. Moreover, with more sophisticated power allocation schemes, Cell-Free systems can be much more energy efficient (bits per Joule) than more conventional cellular deployments, including the power consumption of the fronthaul links. As an example, studies have shown that Cell-Free systems are capable of reaching in excess of 80 Mb/joule in urban conditions and 40 Mb/joule in rural situations, essentially doubling the energy efficiency of conventional single-cell Massive MIMO networks [16].

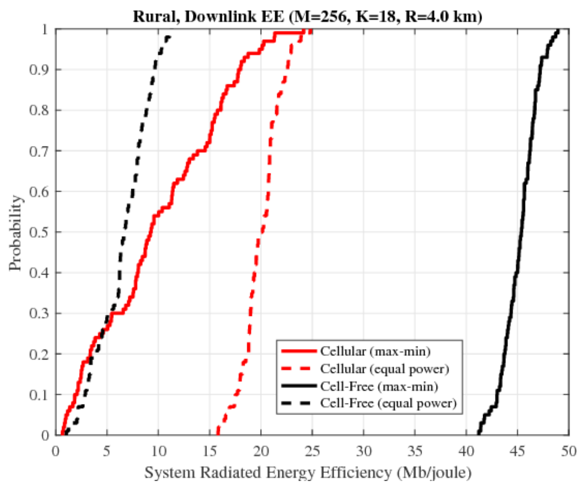

Fig. 6. Cell-Free Massive MIMO and conventional Small-Cell systems are compared using the cumulative distribution of the per-user throughput. [15].

Practical Infrastructure Challenges: Finite Fronthaul

The major bottleneck in the implementation of Cell-Free Massive MIMO is the infrastructure that would be needed to interface with the distributed APs to the CPU. In practice, the capacity of fronthaul links is finite, and thus it can be a limiting factor to the performance improvements of theoretical models. To overcome this, the system should use effective signal quantization and compression methods to make sure that the synchronization and data transfer between the APs and the CPU does not surpass the amount of bandwidth available [17].

Scalability for Real-World Deployment

To scale Cell-Free systems to large cities, it is impractical to have all APs in the network serving all users. A scalable architecture means that only a proportion of the APs around a user need to serve it, and signal processing needs to be partially decentralized. Local partial zero-forcing (PZF) or Protective Combining techniques can be used to ensure that the Complexity load at the CPU is relatively small enough that the network can be scaled without a corresponding exponential overhead growth [18].

VIII. Challenges

One of the key limitations in achieving the theoretical capacity limits of large-scale antenna arrays is the so-called phenomenon of pilot contamination. The number of strictly orthogonal pilot sequences is physically limited because of the short length of the interval of coherence of the channel. As a result, nearby cells are compelled to share the same group of training sequences, which leads to the base station obtaining a superposition of pilots used by users in its own cell and interfering users in neighboring cells. This non-orthogonal reuse generates a correlated channel estimation error, which causes directed inter-cell interference in data transmission which cannot be suppressed simply by adding additional antennas and thus saturates the achievable signal-to-interference-plus-noise ratio (SINR) [19]. The other major implementation challenge is the duplexing schemes of obtaining precise Channel State Information (CSI). Although Time-Division Duplexing (TDD) is based on the channel reciprocity to estimate the downlink channel using uplink pilots without depending on the size of the base station array, the application of this technology in Frequency-Division Duplexing (FDD) systems poses a serious overhead problem. In traditional FDD, the amount of training needed on the downlink is linearly related to the number of base station antennas. This training overhead may entirely absorb the coherence block when a deployment of hundreds of transmitting elements is involved, and there is no zero time-frequency resource available to transmit actual data payloads [20]. More so, the economic viability of the expansion of the base station array determines the utilization of cheap, energy-efficient Radio Frequency (RF) elements. Nevertheless, the exploitation of cheap hardware inevitably brings about such drastic compromises as phase noise, non-linear distortion and low-resolution quantization errors. Also accompanying this is the bottleneck of fronthaul capacity; controlling the huge amount of baseband data being created by hundreds of active transceivers, is an immense burden on the connections between the remote radio heads and the centralized baseband processing units. Therefore, it is a fundamental challenge to design distributed signal processing architectures that support hardware imperfections and achieve hard constraints on fronthaul bandwidth to enable real-world network deployment [20].

IX. Conclusion

The paper has presented an extensive discussion of Massive MIMO systems which are imperative to improve Spectral Efficiency (SE) and Energy Efficiency (EE) of 5G and beyond networks. We compared different linear signal processing algorithms and new deep learning models, outlining the critical trade-offs between computational complexity and system capacity. Moreover, the work highlighted the paradigm shift to Cell-Free Massive MIMO architectures to address the shortcomings of conventional cell-edge designs and evenly distribute services. Nonetheless, a real-world implementation is still impeded by physical layer issues (including pilot contamination, limited fronthaul capacity, and hardware impairments). The future needs to be built upon scalable and decentralized processing architecture and the use of highly developed AI-based methods against these limitations, which will eventually lead to sustainable and ultra-reliable 6G networks.

Appendix

A. Simulation Source Code for Coverage Heatmaps

The following MATLAB script provides the graphical user interface (GUI) and the underlying simulation environment developed to evaluate the spatial coverage of the systems discussed in Section VII. Specifically, the code computes and visualizes the received signal strength (in dB) across a defined two-dimensional grid, offering a direct comparison between a traditional co-located Cellular Massive MIMO topology and a distributed Cell-Free Massive MIMO architecture.

The simulation permits the dynamic adjustment of fundamental network parameters, including the total simulation area, the number of Access Points/Antennas (M), and the total transmit power (P_t). A deterministic path loss model is implemented using a path loss exponent of 3.5. In the traditional cellular scenario, the total transmit power is entirely concentrated at a single, centrally located base station. Conversely, the Cell-Free scenario uniformly distributes the access points across the grid using a fixed random seed for reproducibility, with each access point transmitting at a scaled power fraction of P_t/M. The generated heatmaps visually corroborate the enhanced macro-diversity and more uniform coverage distribution inherent to the Cell-Free approach.

```
function MassiveMIMO_GUI()
    % ==================================
    % 1. Main Figure Initialization
    % ==================================
    fig = uifigure('Name', 'Massive MIMO
        Coverage Simulator', 'Position',
        [100, 100, 1150, 650]);
    fig.Color = [0.94 0.94 0.94];

    % ==================================
    % 2. PANEL 1: Simulation Parameters
    % ==================================
    pnlParams = uipanel(fig, 'Title', '
        Simulation Parameters', ...
        'Position', [20, 420, 300, 200], '
            FontSize', 12, 'FontWeight', '
            bold', ...
        'BackgroundColor', [0.96 0.96
            0.96], ...
        'BorderType', 'line', 'BorderWidth'
            , 1.5);

    uilabel(pnlParams, 'Text', 'Area Size (
        m):', 'Position', [15, 130, 100,
        22]);
    areaEdit = uieditfield(pnlParams, '
        numeric', 'Position', [120, 130,
        150, 22], 'Value', 1000);

    uilabel(pnlParams, 'Text', 'Number of
        APs:', 'Position', [15, 90, 100,
        22]);
    apsEdit = uieditfield(pnlParams, '
        numeric', 'Position', [120, 90,
        150, 22], 'Value', 64);

    uilabel(pnlParams, 'Text', 'Tx Power (W
        ):', 'Position', [15, 50, 100, 22])
        ;
    powerEdit = uieditfield(pnlParams, '
        numeric', 'Position', [120, 50,
        150, 22], 'Value', 1);

    % Simulation Execution Button
    simBtn = uibutton(pnlParams, 'push', '
        Text', 'Run Simulation', ...
        'Position', [60, 10, 180, 35], '
            BackgroundColor', [0 0.45
            0.74], ...
        'FontColor', 'w', 'FontSize', 14, '
            FontWeight', 'bold', ...
        'ButtonPushedFcn', @(btn,event)
            runSimulation());

    % ===================================
    % 3. PANEL 2: Project & Team Details (
        Command Window Style)
```

```
    % ===================================
    pnlTeam = uipanel(fig, 'Title', 'Project Information', ...
        'Position', [20, 20, 300, 380], 'FontSize', 12, 'FontWeight', 'bold', ...
        'BackgroundColor', [0.96 0.96 0.96], ...
        'BorderType', 'line', 'BorderWidth', 1.5);

    % Formatting text exactly like the Command Window image
    infoText = sprintf([...
        '===========================\n' ...
        'DIGITAL COMMUNICATION PROJECT    \n' ...
        '===========================\n\n' ...
        'Supervised By:\n' ...
        ' -> Dr. Mohammed Hammouda\n' ...
        ' -> Eng. Salma Samy\n\n' ...
        '===============================\n\n' ...
        'Prepared By:\n' ...
        ' -> Kareem Mohammed   (238253)\n' ...
        ' -> Rawan Habib       (235067)\n' ...
        ' -> Rawan Walaa       (209673)\n' ...
        ' -> Mahmoud Tarek     (229210)\n\n' ...
        '===============================']);

    % Using Monospaced font to keep the ASCII borders aligned perfectly
    uitextarea(pnlTeam, 'Value', infoText, 'Position', [10, 10, 280, 335], ...
        'Editable', 'off', 'FontSize', 12, 'FontName', 'Courier New', ...
        'BackgroundColor', [0.15 0.15 0.15], 'FontColor', [0.9 0.9 0.9]); % Dark background for terminal look

    % ==================================
    % 4. PANEL 3: Coverage Heatmaps Axes
    % ==================================
    pnlPlots = uipanel(fig, 'Title', 'Coverage Heatmaps (Signal Strength in dB)', ...
        'Position', [340, 20, 780, 600], 'FontSize', 12, 'FontWeight', 'bold', ...
        'BackgroundColor', [0.96 0.96 0.96], ...
        'BorderType', 'line', 'BorderWidth', 1.5);

    axCellular = uiaxes(pnlPlots, 'Position', [20, 80, 340, 450]);
    title(axCellular, 'Traditional Cellular Coverage');
    box(axCellular, 'on');

    axCellFree = uiaxes(pnlPlots, 'Position', [420, 80, 340, 450]);
    title(axCellFree, 'Cell-Free Distributed Coverage');
    box(axCellFree, 'on');

    % Run simulation automatically upon starting the GUI
    runSimulation();

    % =================================
    % 5. Logic Function: Simulation & Plotting
    % =================================
    function runSimulation()
        % Fetch values from GUI inputs
        area_size = areaEdit.Value;
        M = apsEdit.Value;
        pt = powerEdit.Value;
        alpha_pl = 3.5; % Path loss exponent

        % Generate spatial grid coordinates
        grid_resolution = 100;
        x = linspace(-area_size/2, area_size/2, grid_resolution);
        y = linspace(-area_size/2, area_size/2, grid_resolution);
        [X, Y] = meshgrid(x, y);

        % --------------------------------
        % Scenario A: Cellular Calculation
        % --------------------------------
        dist_cellular = sqrt(X.^2 + Y.^2);
        dist_cellular(dist_cellular < 1) = 1;

        Rx_Cellular = (M * pt) ./ (dist_cellular.^alpha_pl);
        Rx_Cellular_dB = 10 * log10(Rx_Cellular);

        % Plotting Cellular
        contourf(axCellular, X, Y, Rx_Cellular_dB, 20, 'LineStyle', 'none');
        colormap(axCellular, 'turbo');
        hold(axCellular, 'on');
        plot(axCellular, 0, 0, 'w^', 'MarkerSize', 12, 'MarkerFaceColor', 'k');
        hold(axCellular, 'off');
        xlabel(axCellular, 'Distance (m)');
        ylabel(axCellular, 'Distance (m)');

        % Add Colorbar and fix limits so Tx Power changes are visible
        cb1 = colorbar(axCellular);
        cb1.Label.String = 'Received Power (dB)';
```

```
        clim(axCellular, [-120 0]); % Fixed
            color limits!

        % ---------------------------------
        % Scenario B: Cell-Free Calculation
        % ---------------------------------
        rng(42); % Fixed seed
        AP_pos = (rand(M, 2) - 0.5) *
            area_size;
        Rx_CellFree = zeros(size(X));

        for i = 1:M
            dist_AP = sqrt((X - AP_pos(i,1)
                ).^2 + (Y - AP_pos(i,2))
                .^2);
            dist_AP(dist_AP < 1) = 1;
            Rx_CellFree = Rx_CellFree + ((
                pt / M) ./ (dist_AP.^
                alpha_pl));
        end
        Rx_CellFree_dB = 10 * log10(
            Rx_CellFree);

        % Plotting Cell-Free
        contourf(axCellFree, X, Y,
            Rx_CellFree_dB, 20, 'LineStyle'
            , 'none');
        colormap(axCellFree, 'turbo');
        hold(axCellFree, 'on');
        plot(axCellFree, AP_pos(:,1),
            AP_pos(:,2), 'w*', 'MarkerSize'
            , 5);
        hold(axCellFree, 'off');
        xlabel(axCellFree, 'Distance (m)');
        ylabel(axCellFree, 'Distance (m)');

        % Add Colorbar and fix limits to
            match Cellular
        cb2 = colorbar(axCellFree);
        cb2.Label.String = 'Received Power
            (dB)';
        clim(axCellFree, [-120 0]); % Fixed
            color limits to match!
    end
end
```

Listing 1. Coverage Heatmap Simulation Code

B. Simulation Parameters

The table below outlines the important parameters employed in the MATLAB simulation environment to create the coverage heatmaps:

TABLE I
SYSTEM SIMULATION PARAMETERS

Parameter	**Value**
Simulation Area Size	1000×1000 m
Number of Access Points (M)	64
Total Transmit Power (P_t)	1 W
Path Loss Exponent (α)	3.5
Grid Resolution	100×100 points
Power Scaling Strategy	P_t/M (Cell-Free)

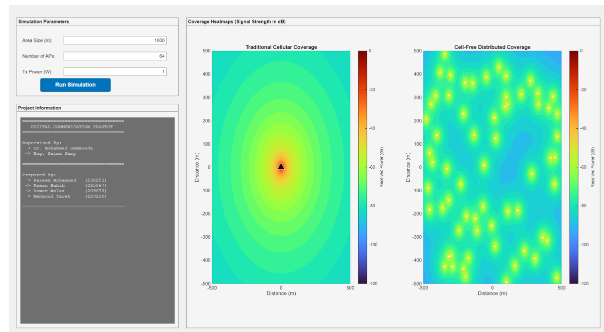

Fig. 7. Coverage Heatmap generated by the MATLAB simulation, comparing Cellular and Cell-Free architectures.

C. Analysis of Heatmap Results

The heatmaps created give a graphical confirmation of the theoretical benefits of Cell-Free Massive MIMO compared to co-located systems. The signal strength in the Traditional Cellular case is very concentrated at the center with a steep fall-off (Path Loss) around the cell edges that forms possible dead zones.

On the other hand, in Cell-Free Distributed heatmap the power distribution is much more uniform throughout the area. Distributing 64 Access Points, the system essentially avoids the distance-dependent path loss, so that users anywhere are assured of a high likelihood of being within proximity of at least one transmitter. It is this increased macro-diversity that is the driving force behind the increased 95%-likely per-user throughput in Section VII.

D. Power Allocation and Signal Model

In the simulation, the transmit power for the Cell-Free architecture follows a uniform allocation strategy where the total power P_t is divided equally among all distributed Access Points. The power transmitted by the m-th AP is given by:

$$P_m = \frac{P_t}{M} \tag{7}$$

This corresponds to the power control coefficient η_{mk} in the received signal model:

$$r_{d,k} = \sqrt{\rho_d} \sum_{m=1}^{M} \sqrt{\eta_{mk}} g_{mk} \hat{g}_{mk}^* q_k + \text{Interference} + \text{Noise} \tag{8}$$

where $\eta_{mk} = 1/M$ ensures that the network operates within its total power budget while providing consistent service quality across the deployment area.

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REFERENCES

[1] L. Lu, G. Y. Li, A. L. Swindlehurst, A. Ashikhmin, and R. Zhang, "An overview of massive mimo: Benefits and challenges," *IEEE Journal on Selected Topics in Signal Processing*, vol. 8, pp. 742–758, 10 2014.

[2] M. A. Albreem, A. H. Habbash, A. M. Abu-Hudrouss, and S. S. Ikki, "Overview of precoding techniques for massive mimo," pp. 60 764–60 801, 4 2021.

[3] M. A. Albreem, M. Juntti, and S. Shahabuddin, "Massive mimo detection techniques: A survey," *IEEE Communications Surveys and Tutorials*, vol. 21, pp. 3109–3132, 10 2019. [Online]. Available: https://ieeexplore.ieee.org/document/8804165

[4] D. C. Araújo, T. Maksymyuk, A. L. de Almeida, T. Maciel, J. C. Mota, and M. Jo, "Massive mimo: Survey and future research topics," *IET Communications*, vol. 10, pp. 1938–1946, 10 2016. [Online]. Available: /doi/pdf/10.1049/iet-com.2015.1091

[5] K. N. V. Prasad, E. Hossain, and V. K. Bhargava, "Energy efficiency in massive mimo-based 5g networks: Opportunities and challenges," *IEEE Wireless Communications*, vol. 24, pp. 86–94, 1 2017. [Online]. Available: https://ieeexplore.ieee.org/document/7811130

[6] T. L. Marzetta, "Massive mimo: An introduction," *Bell Labs Technical Journal*, vol. 20, pp. 11–12, 2015. [Online]. Available: https://ieeexplore.ieee.org/document/7064850

[7] H. Q. Ngo, E. G. Larsson, and T. L. Marzetta, "Energy and spectral efficiency of very large multiuser mimo systems," 5 2012. [Online]. Available: https://arxiv.org/pdf/1112.3810

[8] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, "Massive mimo for next generation wireless systems," *IEEE Communications Magazine*, vol. 52, pp. 186–195, 1 2014. [Online]. Available: http://arxiv.org/abs/1304.6690

[9] E. Björnson, L. Sanguinetti, J. Hoydis, and M. Debbah, "Optimal design of energy-efficient multi-user mimo systems: Is massive mimo the answer?" 3 2015. [Online]. Available: http://arxiv.org/abs/1403.6150

[10] T. L. Marzetta, "Noncooperative cellular wireless with unlimited numbers of base station antennas," *IEEE Transactions on Wireless Communications*, vol. 9, pp. 3590–3600, 11 2010. [Online]. Available: https://ieeexplore.ieee.org/document/5595728

[11] X. Gao, L. Dai, C. Yuen, and Y. Zhang, "Low-complexity mmse signal detection based on richardson method for large-scale mimo systems," *IEEE Vehicular Technology Conference*, 11 2014. [Online]. Available: https://ieeexplore.ieee.org/document/6966041

[12] H. He, C. K. Wen, S. Jin, and G. Y. Li, "A model-driven deep learning network for mimo detection," *2018 IEEE Global Conference on Signal and Information Processing, GlobalSIP 2018 - Proceedings*, pp. 584–588, 7 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8646357

[13] G. Y. Li, Z. Xu, C. Xiong, C. Yang, S. Zhang, Y. Chen, and S. Xu, "Energy-efficient wireless communications: Tutorial, survey, and open issues," *IEEE Wireless Communications*, vol. 18, pp. 28–35, 12 2011. [Online]. Available: https://ieeexplore.ieee.org/document/6108331

[14] H. Ying, D. Xu, B. Ji, Y. Huang, and Y. Luxi, "Energy-efficiency of very large multi-user mimo systems," *2012 International Conference on Wireless Communications and Signal Processing, WCSP 2012*, 2012. [Online]. Available: https://ieeexplore.ieee.org/document/6542888

[15] H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, H. Q. Ngo, A. Ashikhmin, H. Yang, E. G. Larsson, and T. L. Marzetta, "Cell-free massive mimo versus small cells," *IEEE Transactions on Wireless Communications*, vol. 16, p. 1, 2017.

[16] H. Yang and T. L. Marzetta, "Energy efficiency of massive mimo: Cell-free vs. cellular," *IEEE Vehicular Technology Conference*, vol. 2018-June, pp. 1–5, 7 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8417645

[17] P. Parida, H. S. Dhillon, and A. F. Molisch, "Downlink performance analysis of cell-free massive mimo with finite fronthaul capacity," *IEEE Vehicular Technology Conference*, vol. 2018-August, 7 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8690797

[18] E. Bjornson and L. Sanguinetti, "Scalable cell-free massive mimo systems," *IEEE Transactions on Communications*, vol. 68, pp. 4247–4261, 7 2020. [Online]. Available: http://arxiv.org/abs/1908.03119

[19] M. Boulouird, A. Riadi, and M. M. Hassani, "Pilot contamination in multi-cell massive-mimo systems in 5g wireless communications," *Proceedings of 2017 International Conference on Electrical and Information Technologies, ICEIT 2017*, vol. 2018-January, pp. 1–4, 1 2018. [Online]. Available: https://ieeexplore.ieee.org/document/8255299

[20] E. Björnson, E. G. Larsson, and T. L. Marzetta, "Massive mimo: Ten myths and one critical question," *IEEE Communications Magazine*, vol. 54, pp. 114–123, 2 2016. [Online]. Available: http://arxiv.org/abs/1503.06854